Simulation and Sensitivity Analysis of Chaotic Systems

Task 1 Report

July 16, 2026

Part A — Discrete Chaotic Systems

A.1 Logistic Map

$$x_{n+1} = r\,x_n(1 - x_n) \tag{1}$$

Setting	Value
State variable	$x \in [0, 1]$
Control parameter	r
Initial condition	$x_0 = 0.5$
Iterations (time series)	1000, transient = 300 discarded
Demo parameter	$r = 3.9$ (chaotic)
Bifurcation sweep	$r \in [2.5, 4.0]$, step = 0.002; 1000 iterations/point, transient = 500

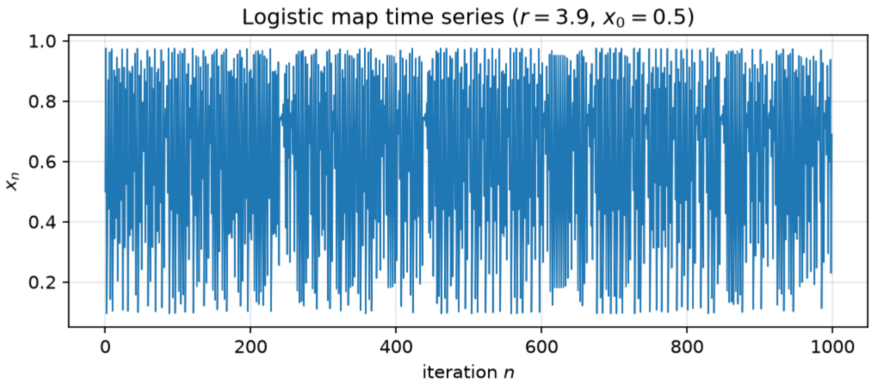


Figure 1: Logistic map time series at $r = 3.9$: the trajectory never settles, jumping irregularly across $[0, 1]$.

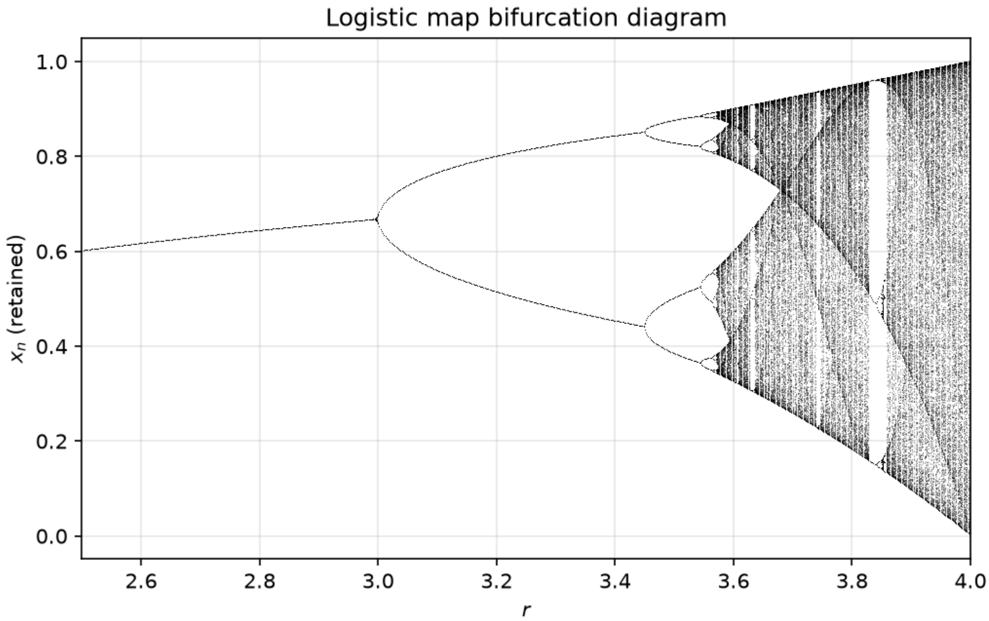


Figure 2: Logistic map bifurcation diagram.

Observation. For $r \lesssim 3.0$ the system settles to a single fixed point; increasing r triggers a period-doubling cascade ($1 \to 2 \to 4 \to \dots$) that accumulates at $r \approx 3.57$, beyond which the map is chaotic almost everywhere. Narrow bright bands of periodicity (e.g. the period-3 window near $r \approx 3.83$) reappear inside the chaotic region, showing that chaos and order are interleaved rather than strictly separated.

A.2 Hénon Map

$$x_{n+1} = 1 - ax_n^2 + y_n, \qquad y_{n+1} = bx_n \tag{2}$$

Setting	Value
State variables	(x, y)
Control parameters	a (nonlinearity), b (contraction)
Classic chaotic parameters	$a = 1.4$, $b = 0.3$
Initial condition	$(x_0, y_0) = (0, 0)$
Iterations (attractor)	5000, transient = 500 discarded
Bifurcation sweep	$a \in [1.0, 1.4]$, step = 0.002, $b = 0.3$ fixed; 1000 iterations/point, transient = 500

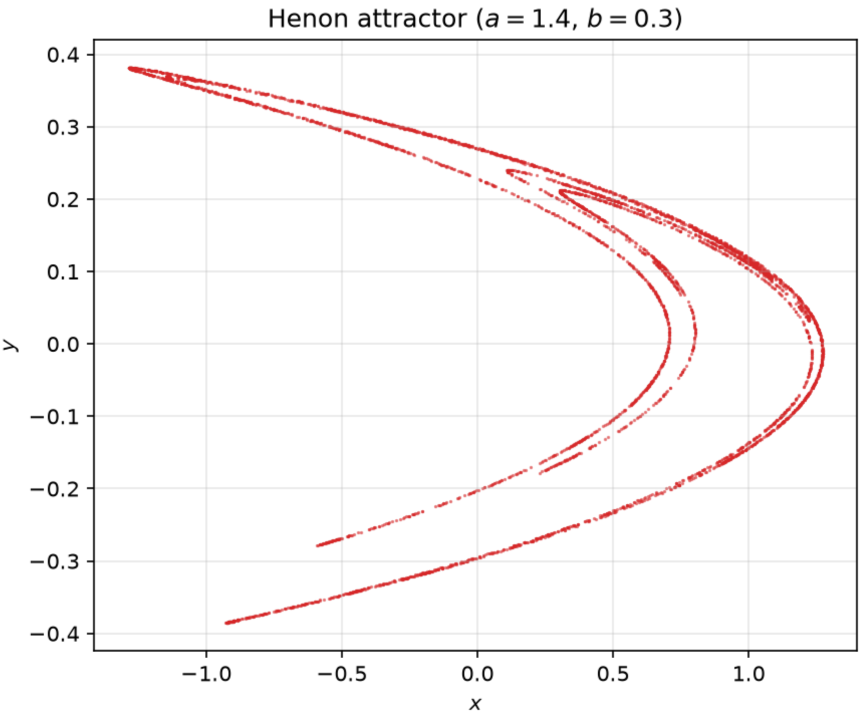


Figure 3: Hénon attractor at $a = 1.4$, $b = 0.3$: the characteristic folded, fractal band structure.

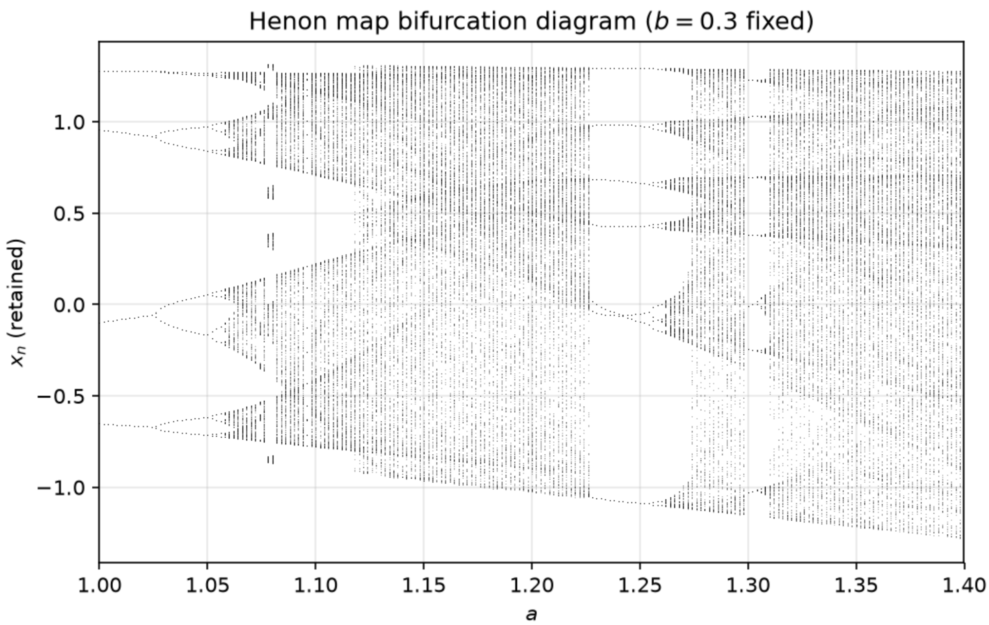


Figure 4: Hénon map bifurcation diagram (sweeping a, $b = 0.3$ fixed).

Observation. As in the logistic map, the Hénon system moves from a stable fixed point through period-doubling into chaos as a increases, but because the state is two-dimensional the retained x_n values trace out thickened, self-similar bands rather than single lines — a discrete echo of the fractal structure visible directly in the attractor plot at $a = 1.4$.

Part B — Continuous Chaotic and Hyperchaotic Flows

B.1 Lorenz System

$$\dot{x} = \sigma(y - x), \qquad \dot{y} = x(\rho - z) - y, \qquad \dot{z} = xy - \beta z \tag{3}$$

Setting	Value
Parameters	$\sigma = 10$, $\rho = 28$, $\beta = 8/3$
Initial condition	$(1, 1, 1)$
Time step	$dt = 0.01$
Length	$T = 50$
Solver	adaptive RK45 (`solve_ivp`, rtol = atol = 10^{-9})

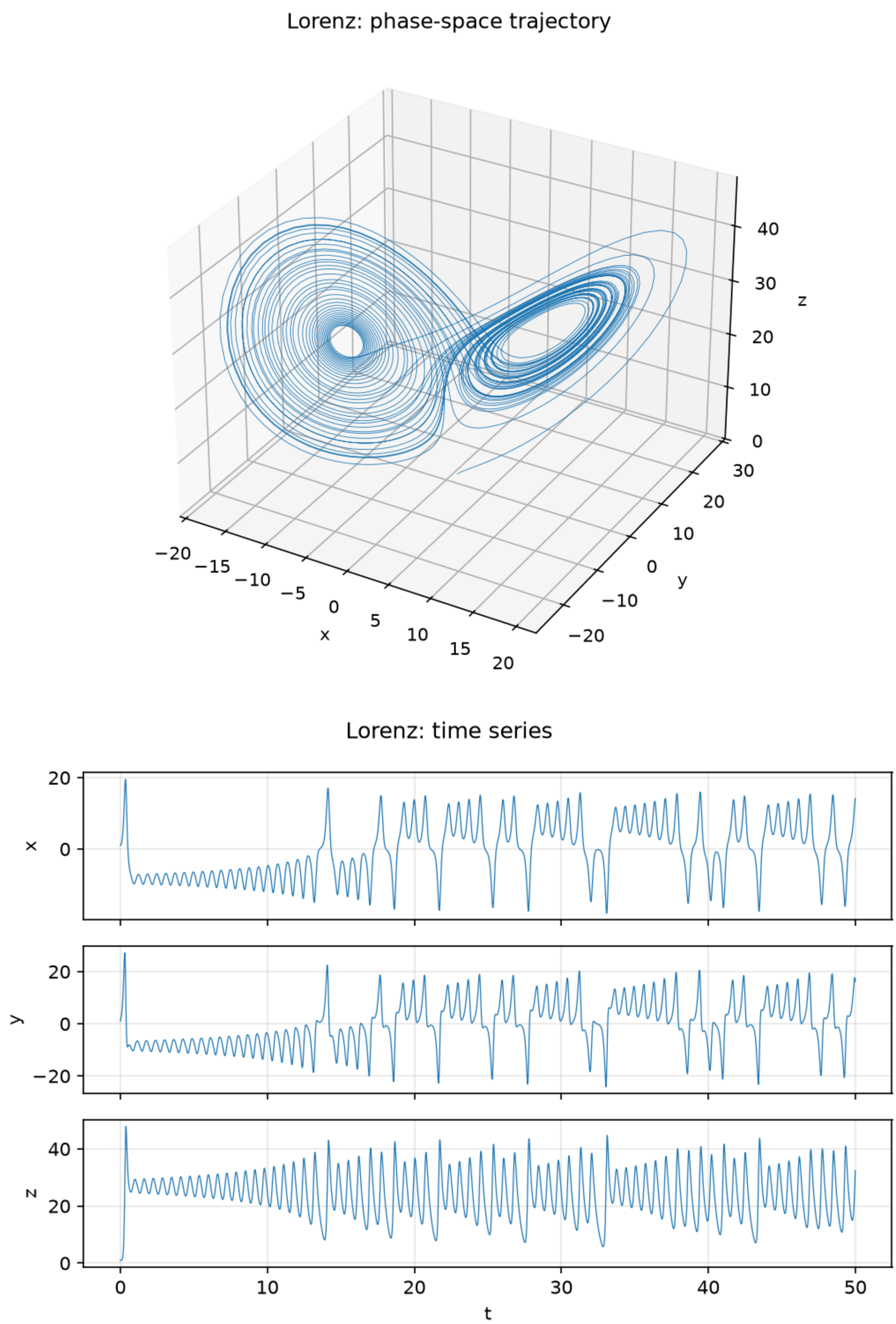
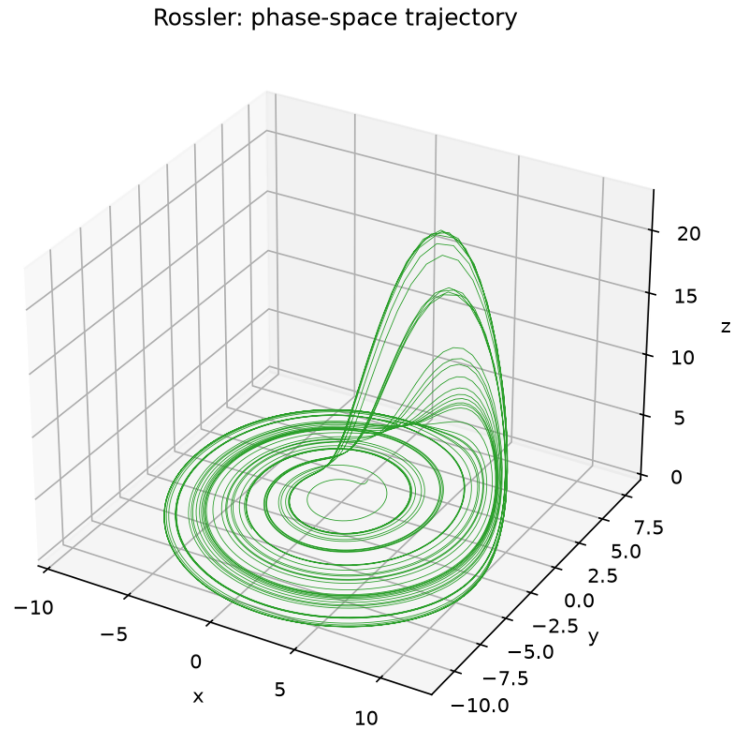


Figure 5: Lorenz system: 3-D phase-space trajectory (the classic butterfly attractor) and $x(t), y(t), z(t)$ time series.

B.2 Rössler System

$$\dot{x} = -y - z, \qquad \dot{y} = x + ay, \qquad \dot{z} = b + z(x - c) \tag{4}$$

Setting	Value
Parameters	$a = 0.2$, $b = 0.2$, $c = 5.7$
Initial condition	$(1, 1, 1)$
Time step	$dt = 0.01$
Length	$T = 250$
Solver	adaptive RK45 (`solve_ivp`)



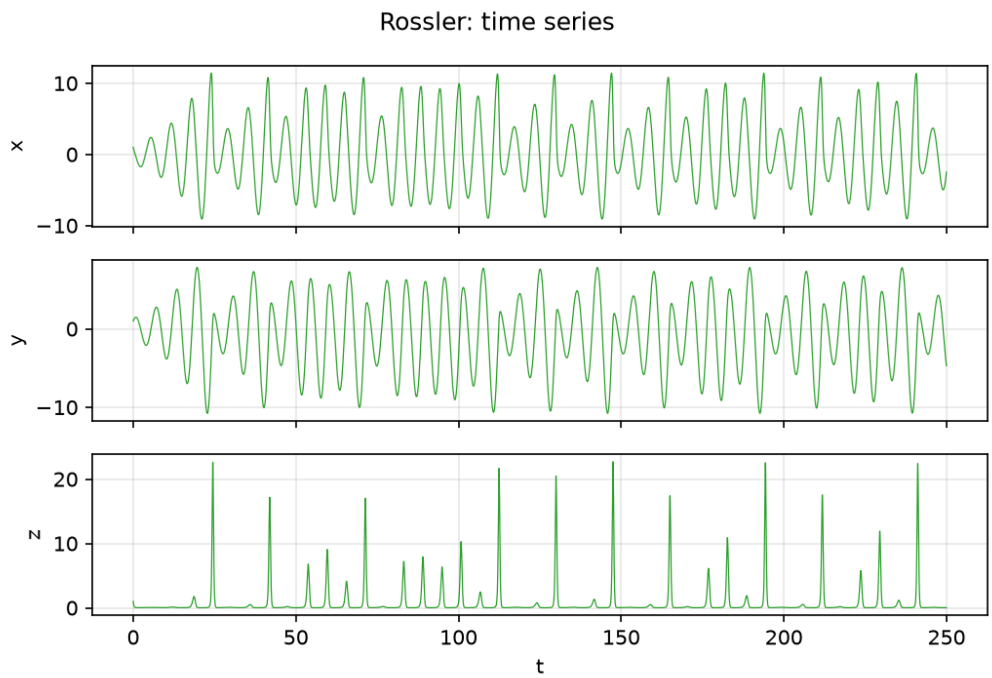
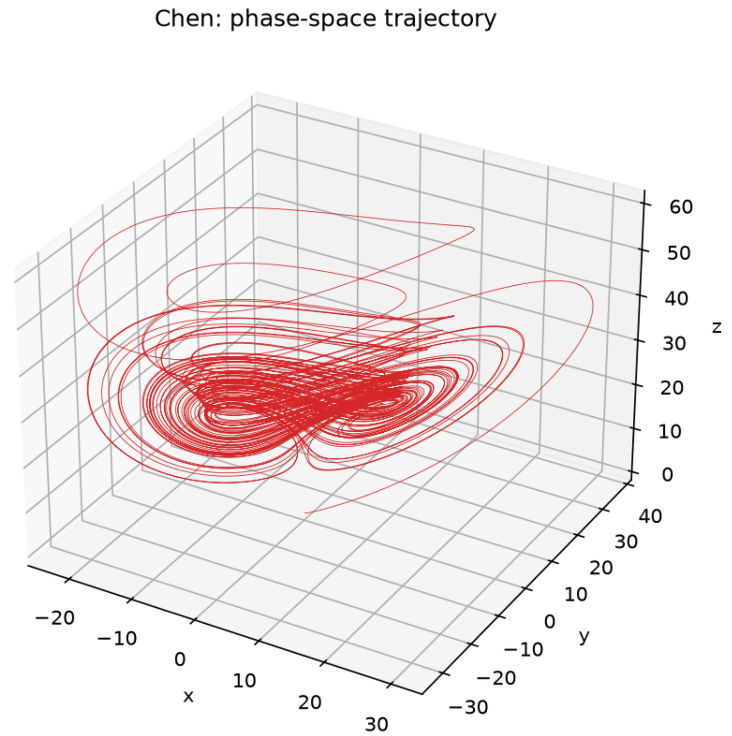


Figure 6: Rössler system: single-scroll spiral attractor and time series.

B.3 Chen System

$$\dot{x} = a(y - x), \qquad \dot{y} = (c - a)x - xz + cy, \qquad \dot{z} = xy - bz \tag{5}$$

Setting	Value
Parameters	$a = 35$, $b = 3$, $c = 28$
Initial condition	$(-0.1, 0.5, -0.6)$
Time step	$dt = 0.002$ (Chen is stiffer than Lorenz)
Length	$T = 50$
Solver	adaptive RK45 (`solve_ivp`, tight tolerances)



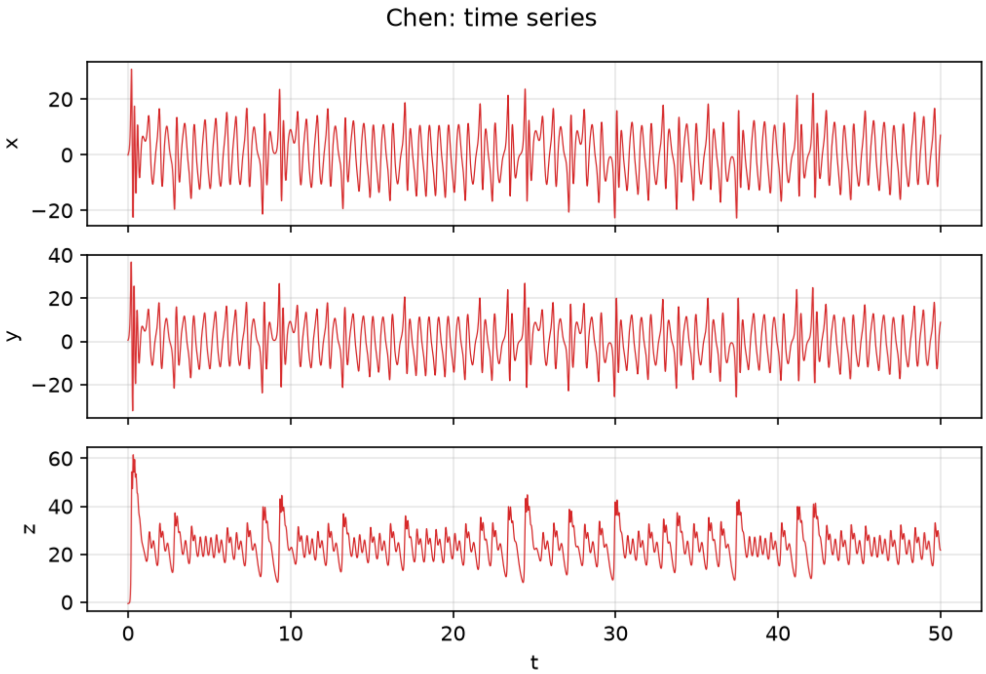


Figure 7: Chen system: double-scroll attractor and time series.

Lorenz vs. Chen. Both attractors have two lobes, but Chen's scrolls are more tightly wound and the switching between lobes is faster and less regular than Lorenz's — consistent with Chen's larger parameters and stiffer dynamics.

B.4 Hyperchaotic Rössler System

$$\dot{x} = -y - z, \qquad \dot{y} = x + ay + w, \qquad \dot{z} = b + xz, \qquad \dot{w} = -cz + dw \tag{6}$$

Setting	Value
Parameters	$a = 0.25$, $b = 3$, $c = 0.5$, $d = 0.05$
Initial condition	$(-10, -6, 0, 10)$
Time step	$dt = 0.01$
Length	$T = 250$
Solver	adaptive RK45 (`solve_ivp`)

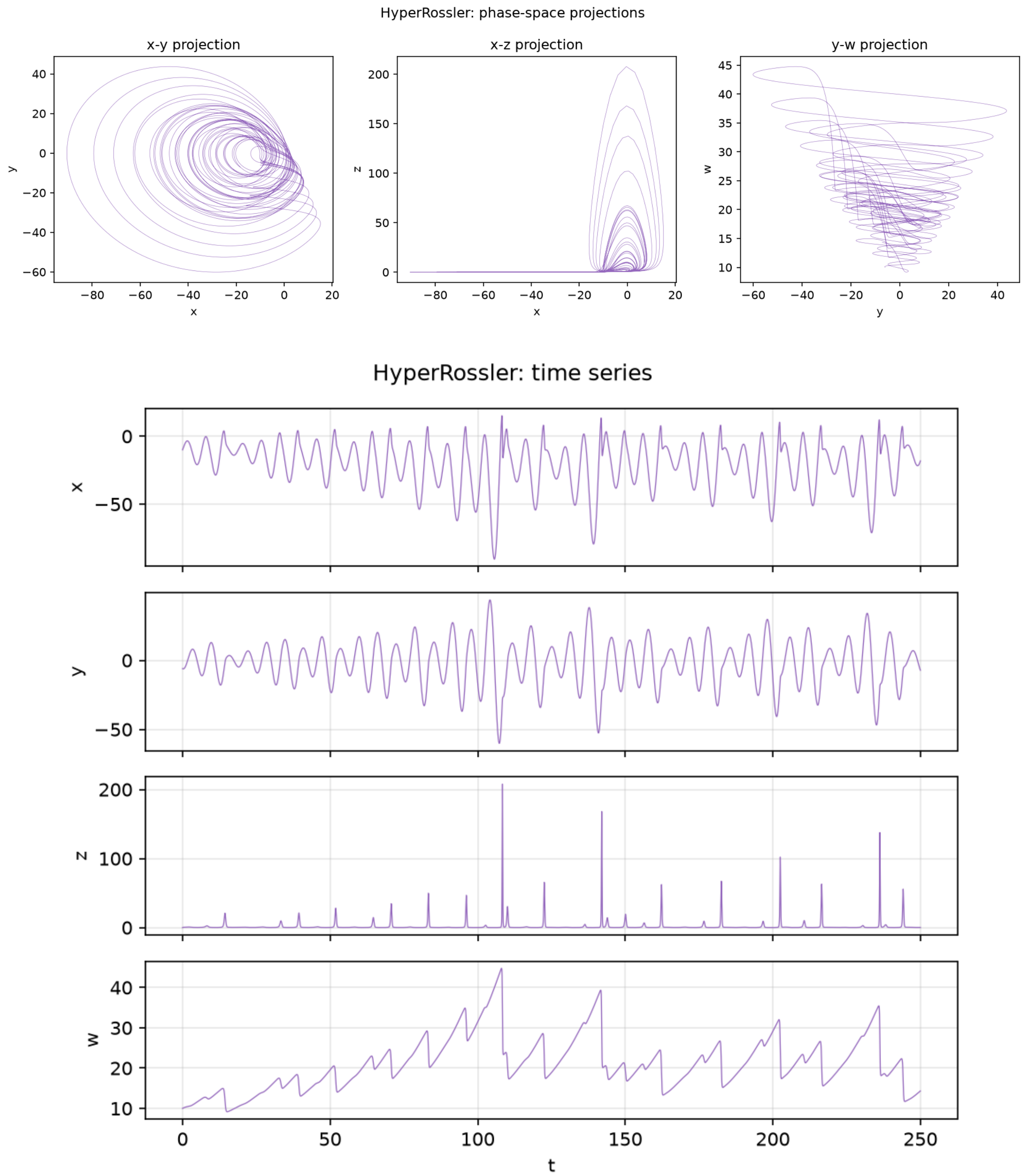


Figure 8: Hyperchaotic Rössler system: 2-D projections of the 4-D attractor (x-y, x-z, y-w) and time series of all four states.

Observation. The coupling of w into $\dot{y}$ injects a second independent stretching direction, so the trajectory folds in more than one plane simultaneously; no single 3-D projection captures the full structure, and the time series of w shows slow oscillations that modulate the faster x, y, z chaos — the qualitative signature of a second positive Lyapunov exponent.

Part C — Sensitivity Analysis

C.1 ρ-sweep of the Lorenz System

Setting	Value
Swept parameter	$\rho \in [0, 30]$, step $= 0.1$ ($\sigma = 10$, $\beta = 8/3$ fixed)
Integration	fixed-step RK4, $dt = 0.01$, $T = 60$, IC $= (1, 1, 1)$
Transient discarded	first 20 time units
Recorded quantity	local maxima of $z(t)$

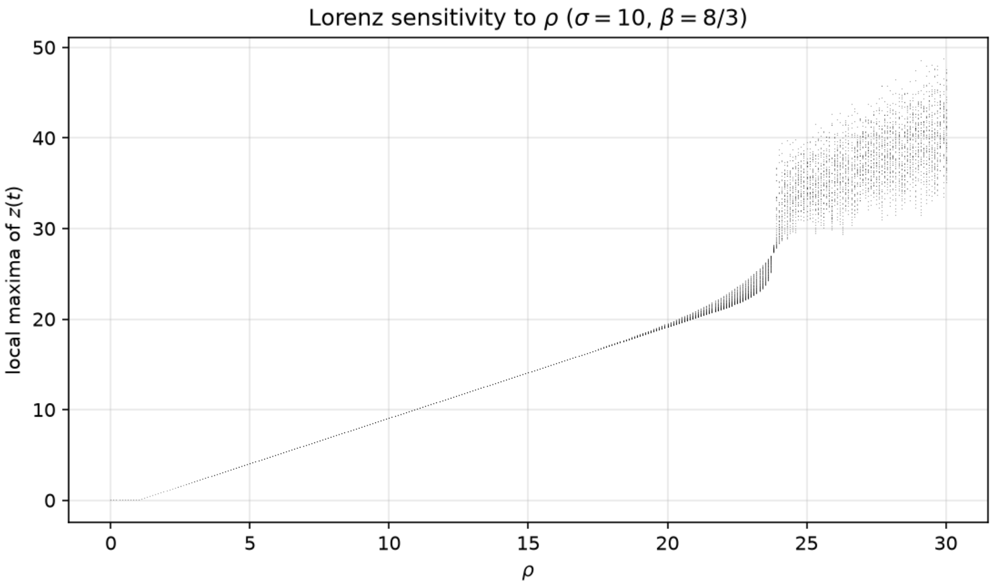


Figure 9: Local maxima of $z(t)$ vs. ρ — the continuous-system analogue of the discrete bifurcation diagrams in Part A.

Discussion. For $\rho \lesssim 24.7$ the trajectory converges to one of the two symmetric fixed points, so a single z-value is recorded per ρ (or none, since a true fixed point has no local maxima — the settled value is plotted instead). Just past $\rho \approx 24.7$ the fixed points lose stability and the recorded maxima explode into a dense chaotic band, occasionally interrupted by narrow periodic windows. At the classic value $\rho = 28$ the system sits deep in this chaotic band, matching the attractor seen in Part B.1.

C.2 Step-size Sensitivity

Setting	Value
Parameters	$\sigma = 10$, $\rho = 28$, $\beta = 8/3$; IC $= (1, 1, 1)$; $T = 50$
Step sizes tested	$dt = 0.001$ (fine), 0.01 (baseline), 0.05 (coarse)
Integrator	fixed-step RK4 (same for all three, only dt changed)

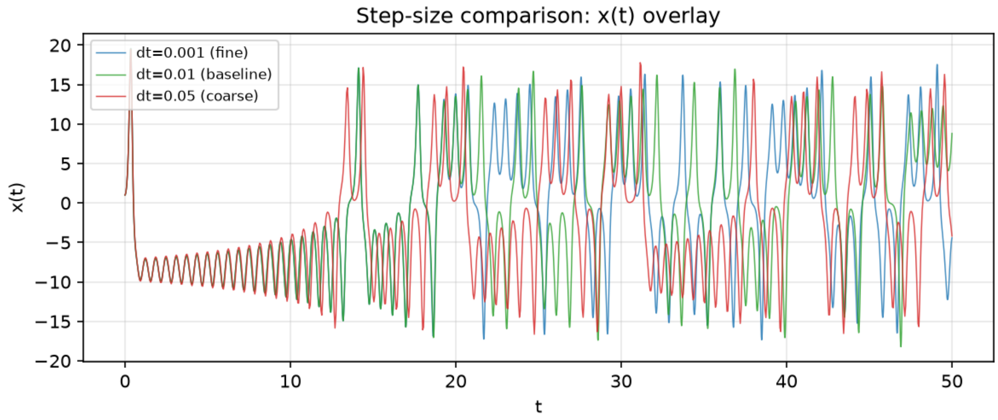


Figure 10: Overlay of $x(t)$ for the three step sizes.

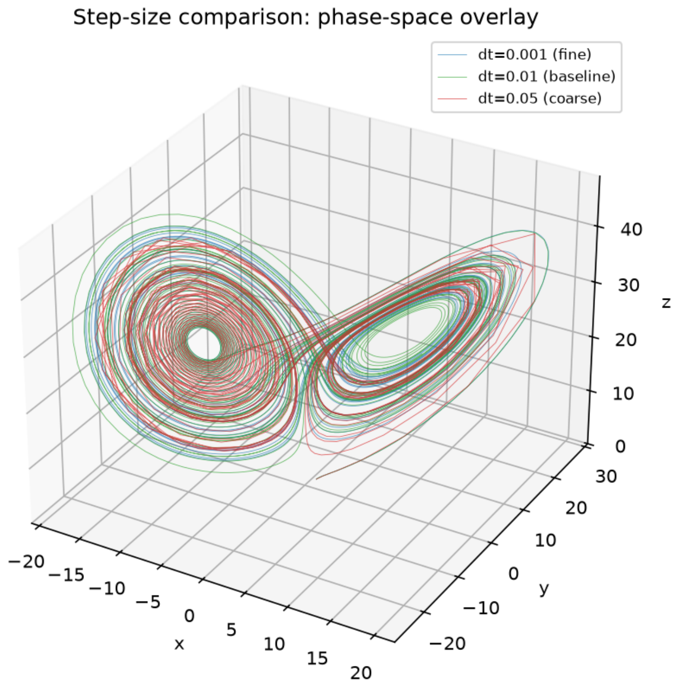


Figure 11: Overlay of the 3-D phase-space trajectories for the three step sizes.

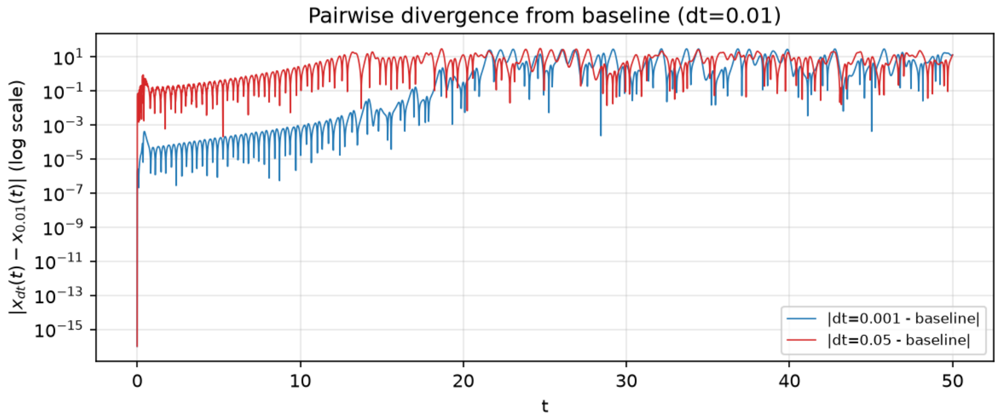


Figure 12: Pairwise divergence $|x_{dt}(t) - x_{0.01}(t)|$ vs. t (log scale), relative to the $dt = 0.01$ baseline.

Concluding Remarks

Discrete vs. continuous chaos. Both types of systems can reach chaos through period-doubling. Discrete maps evolve by iterations, whereas continuous systems evolve in time through ODEs and require numerical integration, making their results sensitive to the chosen step size.

Chaotic vs. hyperchaotic. Chaotic systems have one positive Lyapunov exponent, while hyperchaotic systems have at least two. The additional unstable direction makes hyperchaotic motion more complex and requires multiple projections to represent its structure clearly.